

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 31

Create Referenced Security Groups in a CloudFormation VPC

Amazon VPC | Security Groups | SG References | NFS | CloudFormation

AUTHOR	Michael Salazar
ENVIRONMENT	Personal AWS Lab
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STATUS	Complete

Executive Summary

Logged into **TestAccount1**, recreated the **CloudSLAW** VPC using the CloudFormation template from the previous lab, created a **Database** security group with two inbound rules and one outbound rule, then created a **Backup** security group whose inbound rule references the Database security group and allows NFS traffic on port **2049**. After validating the security group relationship, the created security groups and VPC were deleted to keep lab costs minimal.

CONTROL PATTERN

The important technical point is the security group reference. Backup does not trust a subnet CIDR or a hard-coded IP range for NFS. It trusts resources that have the Database security group attached.

Business Problem

The VPC from the previous labs created network routes and subnet tiers, but workload communication still needs resource-level controls. Security groups provide those controls at the network interface/resource level. This lab demonstrates both CIDR-based rules and security group reference rules, which are foundational for application-tier and database-tier segmentation.

Objectives

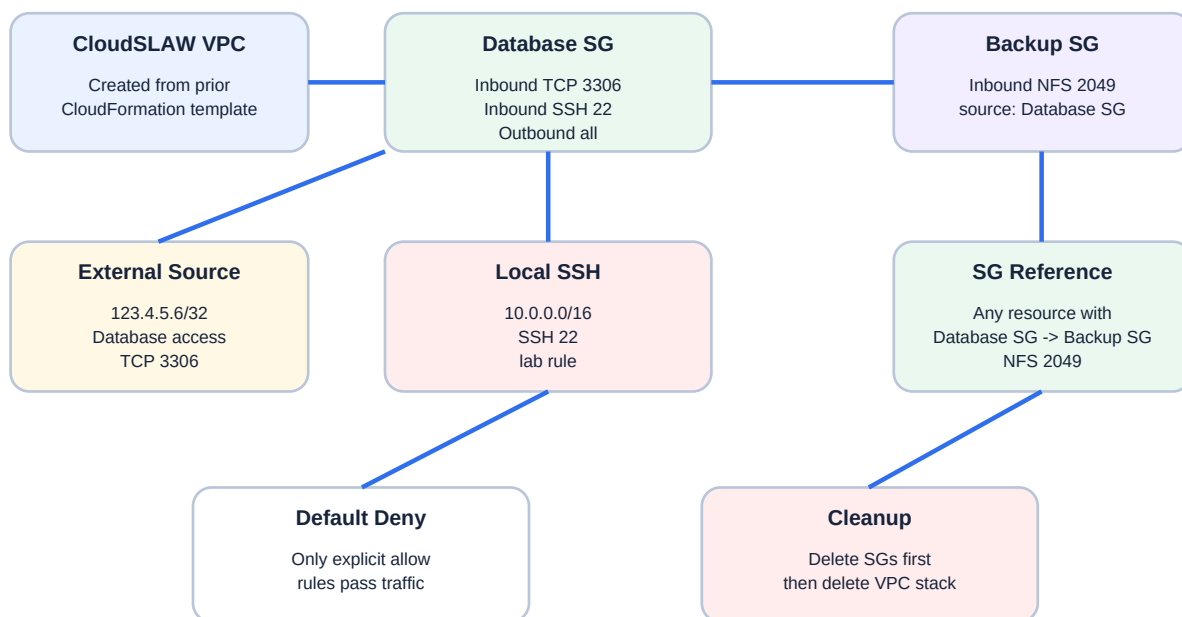
- Use the previously authored CloudFormation template to recreate the CloudSLAW VPC in TestAccount1.
- Create a Database security group in the CloudSLAW VPC.
- Add a custom TCP 3306 inbound rule for database access from a specific external /32 source.
- Add an SSH 22 inbound rule for local VPC access.
- Keep the default outbound rule allowing all outbound traffic for the lab.
- Create a Backup security group in the same CloudSLAW VPC.
- Add an inbound NFS rule on TCP 2049 with source set to the Database security group.
- Delete the lab-created security groups and VPC/stack after testing to avoid unnecessary cost.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Service	Amazon VPC Security Groups
Supporting Service	AWS CloudFormation
Account Context	TestAccount1 / AdministratorAccess
VPC	CloudSLAW created from prior CloudFormation template

Database Security Group	Database
Database Inbound Rule 1	Custom TCP, port 3306, source 123.4.5.6/32, description Database Access
Database Inbound Rule 2	SSH, TCP 22, local VPC source redacted in evidence, description Local SSH
Database Outbound Rule	All traffic to 0.0.0.0/0
Backup Security Group	Backup
Backup Inbound Rule	NFS, TCP 2049, source references Database security group
Backup Outbound Rule	All traffic to 0.0.0.0/0
Cleanup	Deleted created security groups and VPC/CloudFormation stack after the lab

Architecture



Security groups enforce resource-level allow rules and can reference other security groups instead of IP ranges.

Figure 1 - Security group control pattern using CIDR sources and a security-group reference.

Implementation Summary

1. Signed into TestAccount1 using AdministratorAccess.
2. Recreated the CloudSLAW VPC using the CloudFormation template built in Project 30.
3. Opened VPC Security Groups and created a new security group named Database.

4. Selected the CloudSLAW VPC for the Database security group.
5. Added inbound Custom TCP port 3306 from 123.4.5.6/32 with description Database Access.
6. Added inbound SSH port 22 from the local VPC source with description Local SSH.
7. Left the outbound rule as All traffic to 0.0.0.0/0 for the lab.
8. Created a second security group named Backup in the CloudSLAW VPC.
9. Added inbound NFS TCP port 2049 to Backup and set the source to the Database security group.
10. Deleted the created security groups, then deleted the VPC/CloudFormation resources for cost control.

Rule Design

Database	TCP 3306 from 123.4.5.6/32	Allows database access from one specific external source used for the lab.
Database	SSH 22 from local VPC source	Allows local administrative access for the lab scenario.
Database	All outbound to 0.0.0.0/0	Keeps default outbound behavior for the lab.
Backup	NFS TCP 2049 from Database security group	Allows resources assigned to Database SG to reach Backup resources over NFS.
Backup	All outbound to 0.0.0.0/0	Keeps default outbound behavior for the lab.

Evidence

Figure 2 - Database Security Group Created with Database and SSH Rules

The screenshot shows the 'Create security group' page in the AWS Management Console. The 'Basic details' section has 'Security group name' set to 'Database', 'Description' set to 'Database Security group', and 'VPC' set to 'vp-CloudSLAW'. The 'Inbound rules' section contains two rules: a Custom TCP rule for port 3306 with source '123.4.5.6/32' and description 'Database Access', and an SSH rule for port 22 with source '123.4.5.6/32' and description 'Local SSH'. The 'Outbound rules' section has a default rule for 'All traffic' to '0.0.0.0/0'. A warning message at the bottom states: 'Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.'

The Database security group evidence shows name Database, description Database Security group, CloudSLAW VPC selection, inbound Custom TCP port 3306 from 123.4.5.6/32 with description Database Access, inbound SSH TCP port 22 with description Local SSH, and default outbound All traffic to 0.0.0.0/0. Sensitive values were redacted.

Figure 3 - Backup Security Group References Database Security Group for NFS

The screenshot shows the 'Create security group' page in the AWS Management Console. The 'Basic details' section has 'Security group name' set to 'Backup', 'Description' set to 'Database Sends Backup', and 'VPC' set to 'vp-CloudSLAW'. The 'Inbound rules' section contains one rule: an NFS rule for port 2049 with source 'sg-' followed by a redacted security group ID and description 'Database Sends Backup'. The 'Outbound rules' section has a default rule for 'All traffic' to '0.0.0.0/0'. A warning message at the bottom states: 'Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.'

The Backup security group evidence shows name Backup, description Database Sends Backup, CloudSLAW VPC selection, inbound NFS TCP port 2049, and a custom source field referencing a security group. The source identifiers are redacted, but the configuration demonstrates the security group reference pattern.

Security Analysis

Security groups are default-deny controls. Traffic is denied unless an inbound or outbound allow rule permits it.

The Database group demonstrates CIDR-based sources: one narrow /32 source for database access and one local network source for SSH in the lab.

The Backup group demonstrates identity-like network control through a security group reference. The allowed source is not a subnet or IP address; it is membership in the Database security group.

The NFS 2049 rule allows only Database-group resources to talk to Backup-group resources on that port. This is useful for application-to-storage or service-to-service access patterns.

The outbound 0.0.0.0/0 rule is acceptable for this introductory lab, but production designs should review and restrict outbound traffic when appropriate.

Cleanup matters because the VPC was recreated from a stack with NAT-related resources. Deleting the VPC/stack prevents leftover lab resources and potential charges.

Lessons Learned

Security groups are policies in the VPC fabric, not traditional firewall appliances.

Security group rules use source, protocol, and port; they allow matching traffic and implicitly deny everything else.

Security groups can reference other security groups, which enables workload-tier access without hard-coded IP dependencies.

Security group references also affect deletion order. If Backup references Database, delete or remove the referencing Backup rule before deleting Database.

Routes and security groups work together. Security groups cannot allow traffic if the route table does not provide a path.

Production Improvements

- Replace lab IP ranges with approved corporate CIDR blocks or VPN/private connectivity.
- Avoid broad SSH access; use AWS Systems Manager Session Manager where possible.
- Restrict outbound rules instead of leaving all egress open by default.
- Use naming and tagging standards such as app, environment, owner, and data classification.
- Document every security group rule with purpose, ticket/change reference, and expected owner.
- Manage security groups with CloudFormation or Terraform instead of console-created rules.
- Monitor security group changes through CloudTrail, EventBridge, and Security Hub controls.

Skills Demonstrated

Amazon VPC	Security groups	CloudFormation stack reuse
CIDR-based rules	Security group references	NFS / TCP 2049
Database-tier access	Resource-level firewalling	Cleanup discipline

Resume Bullet

RESUME

Configured AWS VPC security groups in a CloudFormation-created VPC, including database access rules, SSH lab access, and a Backup security group that references the Database security group for NFS traffic on TCP 2049, then removed lab resources for cost control.

STAR Interview Story

Situation: The CloudSLAW VPC could be recreated with CloudFormation, but workload communication still needed resource-level network controls.

Task: Create security groups that demonstrate both CIDR-based access and security-group-referenced access, then clean up the lab.

Action: I recreated the VPC from the CloudFormation template, created Database and Backup security groups, configured database and SSH rules, allowed NFS 2049 from Database to Backup using a security group reference, and deleted the security groups and VPC after testing.

Result: The lab demonstrated default-deny security group behavior, source-based allow rules, security group references, and safe cleanup in a repeatable VPC environment.

Version History

v1.0.0	July 22, 2026	Initial documented implementation. Recreated CloudSLAW VPC from template, created Database and Backup security groups, configured database, SSH, outbound, and NFS security-group-reference rules, documented cleanup, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Harness the Magic of Security Groups.
- User-provided Database security group evidence screenshot.
- User-provided Backup security group evidence screenshot.
- Project 30 - Recreate the CloudSLAW VPC with CloudFormation IaC.